SONIA REILLY

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OVERVIEW

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Applied mathematician with expertise in numerical analysis, scientific computing, and statistics. Recent research areas include inverse problems, uncertainty quantification, and optimal experimental design.

EDUCATION

Courant Institute of Mathematical Sciences, New York University — New York, NY
Ph.D. in Mathematics — *Expected 2027*
Thesis Advisor: Georg Stadler
Thesis Title: Scalable Hyperparameter Marginalization in PDE-Governed Bayesian Inverse Problems
Supported by Department of Energy Computational Science Graduate Fellowship

Massachusetts Institute of Technology — Cambridge, MA
S.B. in Mathematics with Computer Science — *2021*

RESEARCH EXPERIENCE

Courant Institute of Mathematical Sciences — New York, NY
Graduate Researcher, advised by Georg Stadler — *2022 – present*

Optimal Experimental Design under Hyperparameter Uncertainty

- Joint work with Noemi Petra, UC Merced
- Optimizing sensor placement to maximize information gain in hierarchical Bayesian inverse problems. Extending my marginalization work to share computation over hyperparameters and designs.

Bridging the Gap between Spatial Statistics and PDE-Governed Inverse Problems

- Joint work with Håvard Rue and Lisa Gaedke-Merzhäuser, KAUST
- Connecting hierarchical Bayesian inverse problems governed by PDEs to integrated nested Laplace approximations (INLA), a marginalization method widely used in spatial statistics.
- Extending my amortized marginalization method to non-Gaussian noise models, such as Poisson count data, using INLA's Laplace approximations.

Amortized Methods for Hierarchical Bayesian Inverse Problems

- Developed scalable deterministic methods of marginalizing hyperparameters in linear Bayesian PDE inverse problems with Gaussian priors.
- Extended state-of-the-art low-rank approximation methods to enable fast computation in problems with nonlinear prior hyperparameters, such as correlation lengths.
- Demonstrated speed and accuracy on a time-dependent 3D advection-diffusion benchmark problem.

MCMC Sampling for Infinite-Dimensional Bayesian Inverse Problems

- Implemented a gradient-free ensemble MCMC sampler for the high-dimensional, concentrated posteriors of infinite-dimensional Bayesian PDE inverse problems, combining affine-invariant methods with the preconditioned Crank–Nicolson method.

Lawrence Berkeley National Laboratory — Berkeley, CA
DOE CSGF Practicum Fellow, advised by Jeff Donatelli — *Summer 2023*

- Designed a method to incorporate measurement uncertainty into the Multi-Tiered Iterative Phasing (M-TIP) algorithm for recovering the density of a molecule imaged using fluctuation X-ray scattering.
- Successfully retrieved densities from images with added constant and variable Gaussian noise.

Johns Hopkins University Applied Physics Laboratory Laurel, MD
Postgraduate Researcher, advised by Sam Yee *Summer 2021*

- Developed a numerical method for the tomographic inversion problem of recovering atmospheric brightness from satellite data which records integrated brightness values.

Undergraduate Researcher, advised by William Swartz *Sep 2016 - Aug 2017, Summer 2018*

- Extracted high-resolution maps of Earth's outgoing radiation from low-resolution satellite data for NASA's first Earth science CubeSat mission, using spherical harmonics and compressive sensing.

Massachusetts Institute of Technology Cambridge, MA
Undergraduate Researcher, advised by Jörn Dunkel *Feb 2020 – Dec 2020*

- Designed coupled-oscillator network topologies that cancel fluctuations in power grid models, and analyzed their resistance to cascading failures.

Undergraduate Researcher, advised by Sai Ravela *Summer 2019*

- Investigated the mathematics of deep learning by modeling the evolution of the weights of neural networks in stochastic gradient descent as a Fokker-Planck system.

PUBLICATIONS

2. **S. Reilly** and G. Stadler, "Amortized Low-Rank Approximation for Hyperparameter Marginalization in PDE-Governed Bayesian Inverse Problems," under review at *SIAM Journal on Scientific Computing*, 2026. arXiv: 2607.03355.

1. W. Swartz, S. Lorentz, S. Papadakis, P. Huang, A. Smith, D. Deglau, Y. Yu, **S. Reilly**, N. Reilly, and D. Anderson, "RAVAN: CubeSat Demonstration for Multi-Point Earth Radiation Budget Measurements," *Remote Sensing*, 11(7):796, 2019. DOI: 10.3390/rs11070796.

INVITED TALKS AND ORGANIZED SESSIONS

3. SIAM Conference on Computational Science and Engineering, Pittsburgh, PA, February 2027 (upcoming; minisymposium co-organizer)
   Session: Sampling-free Bayesian Inference: Bridging Spatial Statistics and PDE-governed Inverse Problems
   Title: "Amortized Hyperparameter Marginalization for Bayesian PDE-governed Inverse Problems"

2. SIAM Annual Meeting 2026, Cleveland, OH, July 2026
   Session: From Theory to Computation: Modern Methods for PDE Inverse Problems
   Title: "Amortized Low-Rank Approximation for Hyperparameter Marginalization in PDE-governed Bayesian Inverse Problems"

1. SIAM Conference on Computational Science and Engineering, Fort Worth, TX, March 2025
   Session: Investigating Inverse Problems using Bayesian Inference: Challenges and Advances
   Title: "Bayesian Inversion of PDE-Governed Problems Using Integrated Nested Laplace Approximations"

CONTRIBUTED TALKS AND POSTERS

5. Numerical Analysis and Scientific Computing RTG Workshop, Rice University, Houston, TX, October 2026

4. ICERM Workshop on Bayesian Inverse Problems and UQ, Providence, RI, March 2026 (poster)

3. IMSI Workshop on Data Assimilation and Inverse Problems for Digital Twins, Chicago, IL, October 2025 (poster)

2. DOE CSGF Annual Program Review Outgoing Fellow Presentation, Washington, D.C., July 2025

1. Mid-Atlantic Numerical Analysis Day, Temple University, Philadelphia, PA, November 2024

AWARDS

SIAM Student Travel Award, SIAM Annual Meeting 2026
SIAM Student Travel Award, SIAM Computational Science and Engineering 2025
Department of Energy Computational Science Graduate Fellowship 2021-2025
NSF Graduate Research Fellowship (Awarded, but declined in favor of DOE CSGF.) 2021
Henry M. MacCracken Fellowship, New York University 2021

TEACHING AND MENTORSHIP

Graduate Teaching Assistant, Courant Institute, NYU

MATH-UA 334: Mathematical Statistics *Spring 2025, 2026 and Fall 2024, 2025, 2026*

- Led two weekly 75-minute recitation sections per semester, each for 30–40 undergraduate juniors and seniors, covering parameter estimation, hypothesis testing, and regression; held weekly office hours.
- Built recitations around guided practice. Opened with a review placing the week's topic in context, followed by worksheets with conceptual questions and open-ended computational problems.
- Wrote over 70 pages of problems, solutions, and lecture reviews (soniareilly.github.io/teaching); led exam review sessions and answered questions on the course discussion board.
- Student evaluations: 4.81/5 across four semesters.

MATH-UA 253: Linear and Nonlinear Optimization *Spring 2024*

- Led weekly 75-minute recitations and held office hours for a class of 45 undergraduate seniors.
- Developed Julia coding tutorials, wrote example problems and solutions for weekly worksheets, and prepared reviews of lecture material. Graded exams and responded to discussion board questions.
- Delivered a 90-minute lecture as a substitute for the course instructor.
- Student evaluations: 4.75/5

Research Mentor, Courant Institute, NYU

- Anchi Lin, NYU undergraduate student, Summer 2026
  Project: Delayed Acceptance MCMC for Bayesian Inverse Problems with Hyperparameters
  Independently designed and supervised. Student subsequently applying to graduate math programs.

Teaching Development

Introduction to Pedagogy, NYU Office of Teaching Excellence and Innovation *January 2026*

- Month-long intensive covering pedagogy research; developed a syllabus for Mathematical Statistics, including course goals and an assessment plan.

SERVICE AND OUTREACH

CAS-GSAS Student Network Mentoring Program, Graduate Mentor 2026
- Mentored an NYU rising senior applying to graduate programs in math.

Courant Association of Math PhD Students, Co-President 2023–2024
- Interfaced with the department, raised funds, and organized visiting students' day.
- Oversaw a revamp of the student-organized website that hosts resources for incoming students.

Courant Diversity, Equity, and Inclusion Reading and Outreach Group, Member 2021–2024
- Led and participated in discussions on diversity and equity in mathematics and graduate education.
- Volunteered at a local high school, assisting in evaluating student math projects.

Association for Women in Math, Courant Organizing Committee Member 2021–2023

- Organized social events for women in the Courant math department, and led an effort to invite more women speakers to department colloquia.

Courant Graduate Student/Postdoc Seminar, Co-organizer 2021–2022

- Invited speakers and organized logistics for a seminar connecting first-year students with advisors.

MIT Undergraduate Society of Women in Math, Peer Mentor 2020–2021

- Mentored a younger undergraduate woman in navigating the math department at MIT.

North American Computational Linguistics Open Competition, Organizing Committee 2017–2021

- Helped organize, contributed problems, served on the jury, and hosted the MIT location of this contest for high school students.

SKILLS

Software	Python, Julia, MATLAB, C/C++, parallel computing: CUDA, OpenMP
Languages	Spanish (fluent); German, Italian, French (proficient in reading)